ADHD Focus Monitor

Complete Research & Build Plan

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"Can an AI-powered adaptive intervention system, delivered through a Chrome Extension using real-time eye tracking, significantly reduce attentional distraction and improve self-reported learning quality in adults with ADHD?"

1. Project Overview

The ADHD Focus Monitor is a Chrome Extension that uses real-time eye tracking and facial cue analysis to passively detect attention loss during study sessions. When distraction is detected, an AI system (Cerebras + LLaMA 4) delivers personalized, escalating interventions to restore focus — without interrupting the learning flow unnecessarily.

The system is built around a complete session lifecycle: a pre-session AI conversation that sets goals and study duration, an active monitoring phase with a live countdown timer and a real-time score, an escalating AI-powered intervention ladder when distraction is detected, and a post-session AI debrief that evaluates learning quality and sends a daily report.

1.1 Core Innovations

- Passive distraction detection via eye tracking — no self-reporting required
- AI interventions that are personalized to the user's own study goal and context
- Escalation ladder that starts with intelligent AI messages and only moves to mechanical actions if AI fails
- Gamified focus score (100 points) that gives users real-time feedback on their attention
- Post-session AI debrief that discusses both learning quality and ADHD symptoms experienced
- Automated daily report sent after each session for longitudinal tracking

1.2 Target Conferences

Conference	Why It Fits	Deadline	Event Date
ASSETS 2026	Accessibility & assistive technology for ADHD/neurodivergent users	24 June 2026	October 2026
IMWUT / UbiComp	Interactive systems, real-time behavioral monitoring	1 November 2026	Late 2026

2. Full Session Lifecycle

Every study session using the ADHD Focus Monitor follows a structured five-phase lifecycle. Each phase has a clear purpose and a defined interaction between the user and the system.

Phase 1 — Pre-Session AI Conversation

Before the session timer starts, the user opens the extension popup and is greeted by the AI chatbot. This conversation serves three purposes: it establishes the user's motivation for studying, it sets the study duration, and it gives the AI enough context to make all future interventions relevant and personal.

The conversation flow is as follows:

AI Message	User Input	Stored As
What are you studying today?	Free text (e.g., 'Calculus — integration by parts')	current_topic
Why is this important to you right now?	Free text (e.g., 'Exam in 3 days, I really need to pass')	study_reason
How long do you want to focus today? (in minutes)	Number (e.g., 45)	session_duration
Great. I'll check in if you lose focus. Ready to start?	User clicks START	session_start_time

All four values are stored in Chrome session storage and passed to the AI in every subsequent intervention during this session.

Phase 2 — Active Session: Timer + Score

Once the user clicks START, two visible elements appear on screen:

- A countdown timer showing the remaining study time (e.g., '43:12 remaining'). The timer is displayed as a small, non-intrusive overlay in the corner of the browser.
- A focus score displayed next to the timer, starting at 100 points. This score decreases in real time whenever distraction is detected or an intervention fails.

Score deduction rules:

Event	Points Deducted	Reason
Distraction detected (DISTRACTED signal)	-1 point	Each attention loss counts
AI intervention sent but user remains distracted after 30 seconds	-1 additional point	Failed recovery attempt
Mechanical intervention triggered (skip, scroll, sound, flash)	-1 additional point per level	Escalation = worse focus

The score is shown live throughout the session. At the end, it becomes the primary focus quality metric for the daily report.

Phase 3 — Distraction Detection (WebGazer.js)

WebGazer.js runs entirely in the browser using the device webcam. No video is recorded or sent anywhere. It outputs a continuous attention signal based on three measurable indicators:

Signal	Normal State	Distraction Indicator	Threshold
Gaze Direction	On screen	Off screen	> 3 consecutive seconds
Blink Rate	15–20 per minute	Irregular or very high	> 25 blinks/min for 60 seconds
Head Movement	Stable, facing forward	Turning or tilting away	Sustained movement > 3 seconds

When any one of these three signals triggers, the system outputs a DISTRACTED event and immediately begins the intervention ladder (Phase 4).

Phase 4 — AI-Integrated Escalating Intervention Ladder

This is the core of the system. Every intervention starts with an AI-generated, personalized message. Only if the AI fails to restore focus does the system escalate to a mechanical action. At every mechanical level, the AI message is ALSO sent alongside the mechanical action — the two are never separated.

Level	Trigger	AI Action	Mechanical Action	Points Lost
1	1st or 2nd distraction	Short question about the content or gentle	None	-1 (distraction)

Level	Trigger	AI Action	Mechanical Action	Points Lost
		reminder of study reason		
2	3rd distraction (AI level 1 failed)	Content summary + motivational message tied to study reason	Video skips back 30 seconds OR page scrolls back	-1 (fail) + -1 (mechanical)
3	4th distraction (level 2 failed)	Acknowledge difficulty, offer a micro-strategy (e.g., 'Try reading just the next paragraph')	Unfamiliar short audio cue plays	-1 (fail) + -1 (mechanical)
4	5th distraction (level 3 failed)	Stronger emotional check-in: 'It seems really hard to focus right now. What's going on?'	Screen border flashes color briefly	-1 (fail) + -1 (mechanical)
5	6th+ distraction (level 4 failed)	Full chatbot conversation popup — open-ended support, may suggest a break	Full overlay chatbot replaces all other interventions	-1 (fail) + -1 (mechanical)

The key design principle: the AI is never removed from the process. Even at the highest escalation level, the mechanical action is always paired with an AI message. This ensures the intervention remains human and contextual, not purely robotic.

Phase 5 — Post-Session AI Debrief + Daily Report

When the timer reaches zero (or the user ends the session manually), the extension enters the post-session phase. This has three sequential steps:

Step 1 — AI Debrief Conversation

The AI initiates a short conversation with the user about the session. This conversation covers two areas:

- Learning quality: 'Did you feel you understood the material well today? What was the hardest part?'
- ADHD symptom check: 'How was your focus overall? Did you feel restless, overwhelmed, or zoned out at any point?'

The AI uses the session data (score, distraction count, intervention log) to make the conversation specific and relevant, not generic.

Step 2 — Learning Quality Rating

After the conversation, the user is asked to rate two things on a 1–10 scale:

- How much do you feel you actually learned today?
- How focused did you feel overall during this session?

Step 3 — Daily Report Generation and Submission

The extension automatically compiles the daily report and either displays it for the user to review or sends it directly to the researcher's data collection endpoint. The report contains:

Field	Source	Example Value
Date	System	2026-04-15
Session duration (minutes)	Timer	43
Study topic	Pre-session input	Calculus — integration by parts
Study reason	Pre-session input	Exam in 3 days
Focus score (out of 100)	Live score tracker	74
Total distractions detected	WebGazer log	12
Interventions sent	Intervention logger	9
Highest escalation level reached	Intervention logger	Level 3
AI refocus success rate	Outcome tracker	67%
Self-rated learning quality (1–10)	Post-session input	7
Self-rated focus quality (1–10)	Post-session input	6
AI debrief summary	AI-generated	User reported feeling restless in the last 15 minutes but understood the core concepts.
Condition (A/B/C)	Random assignment	A (Full System)

3. AI Prompt Design

All AI interactions use the Cerebras API with LLaMA 4. Cerebras is chosen for its sub-second inference speed, which is essential for real-time intervention. The free tier is sufficient for a 14-day experiment with 20 participants.

3.1 Pre-Session Chatbot System Prompt

```
You are a friendly AI study coach for someone with ADHD.
Your job right now is to help the user set up their study session.
Ask them what they are studying, why it matters to them,
and how long they want to focus.
Be warm, brief, and encouraging. Max 2 sentences per message.
Do not give advice yet. Just collect the information.
```

3.2 In-Session Intervention Prompt (sent on each distraction)

```
SYSTEM:
You are an AI focus coach for a person with ADHD.
Be warm, direct, never condescending. Max 2 sentences.
Never repeat the same message twice in a session.

SESSION CONTEXT:
- Topic: {current_topic}
- Why they are studying: {study_reason}
- Content summary so far: {running_summary}
- Distraction count: {distraction_count}
- Current focus score: {score}/100
- Time remaining: {time_remaining} minutes
- Current escalation level: {level}

INSTRUCTIONS BY LEVEL:
Level 1: Ask a short question about the content to re-engage them.
Level 2: Summarize what they covered + connect to their study reason.
Level 3: Acknowledge the struggle, offer a tiny concrete next step.
Level 4: Emotional check-in. Ask what is making focus hard right now.
Level 5: Open conversation. Suggest a break if needed. Be supportive.

Respond ONLY with your message to the user. Nothing else.
```

3.3 Post-Session Debrief Prompt

```
SYSTEM:
You are an AI study coach wrapping up a session with an ADHD user.
Be warm, reflective, and non-judgmental.
```

```
SESSION SUMMARY:
- Topic studied: {current_topic}
- Duration: {actual_duration} minutes
- Final focus score: {score}/100
- Total distractions: {distraction_count}
- Highest intervention level reached: {max_level}
- AI refocus success rate: {ai_success_rate}%

TASK:
1. Start with one genuine observation about their session.
2. Ask them how they feel about what they learned today.
3. Ask if they noticed any ADHD symptoms (restlessness, zoning out,
   racing thoughts, overwhelm).
4. End with one encouraging sentence about their next session.

Keep the full conversation under 6 exchanges total.
```

3.4 Example Interactions

Phase	Situation	AI Message
Pre-session	User opens extension	Hey! What are you studying today?
Pre-session	User says 'calculus'	Got it. Why does this matter to you right now?
Level 1 intervention	1st distraction, studying calculus for exam	Quick check — what was the last thing you just read or watched? Try to recall it before moving on.
Level 2 intervention	3rd distraction, 20 min in	You've covered derivatives and the chain rule so far. Your exam is in 3 days — you've got this. What's the next topic on your list?
Level 3 intervention	4th distraction	Focus seems tough right now, and that's okay. Try this: just read the next heading and the first sentence below it. Nothing more.
Post-session debrief	Score was 68, 10 distractions	You stayed in the session for 43 minutes even when it got hard — that counts. How do you feel about what you actually understood today?

4. Technical Architecture

4.1 Component Overview

Component	Technology	Purpose
Eye tracking	WebGazer.js (browser-native)	Detect gaze direction, blink rate, head movement
AI conversations	Cerebras API + LLaMA 4	Pre-session chat, interventions, post-session debrief
Video skip	HTML5 video.currentTime property	Skip video back 30 seconds on distraction
Page scroll	window.scrollBy() Chrome API	Scroll page back to last position
Audio cue	Web Audio API (browser-native)	Play short unfamiliar sound
Border flash	CSS injection via content script	Flash screen border color for 2 seconds
Score tracker	Chrome storage.session	Live 100-point focus score
Timer	JavaScript setInterval()	Countdown timer displayed as overlay
Session data	Chrome storage.session	Store topic, reason, duration, logs
Daily report	JSON export / fetch() POST	Compile and submit session data

4.2 Content Extraction for AI Context

The AI needs to know what the user is currently studying to make interventions relevant. Content is extracted differently depending on the page type:

Page Type	Extraction Method	Limitation
YouTube video	YouTube captions fetched via DOM (track element)	Requires captions to be enabled on the video
Article / webpage	document.body.innerText — first 500 words extracted	May include navigation text on some sites
PDF in browser	window.getSelection() on visible text	Works only on browser-rendered PDFs
Unknown / other	User manually types topic in pre-session chat	Fallback for any page type

5. Step-by-Step Build Plan

The build is divided into 6 phases. Complete and test each phase fully before starting the next. Do not skip ahead.

Phase 1 — Basic Extension + Camera (Day 1)

Goal: Chrome extension loads, camera activates, WebGazer prints gaze coordinates to console.

1. Create folder adhd-monitor/ with manifest.json and popup.html
2. Load WebGazer from CDN in popup.html
3. Open Chrome > Extensions > Developer Mode > Load Unpacked
4. Click extension icon, confirm webcam activates
5. Confirm WebGazer prints {x, y} gaze coordinates in console every second

Phase 2 — Distraction Detection (Day 2)

Goal: Console prints DISTRACTED reliably when you look away for 3+ seconds.

6. Add gaze threshold: if gaze off-screen for 3 consecutive seconds → DISTRACTED
7. Add blink rate monitor: if blink rate exceeds 25/min for 60 seconds → DISTRACTED
8. Add head movement monitor: if head turns away for 3+ seconds → DISTRACTED
9. Test each signal individually. Confirm no false positives when focused.

Phase 3 — Pre-Session Chat + Timer + Score (Days 3–4)

Goal: Full pre-session flow works end to end before any distraction detection.

10. Build chatbot UI in popup.html — simple text input and message display
11. Connect to Cerebras API. Store API key in chrome.storage.local
12. Implement pre-session conversation: collect topic, reason, duration
13. After user clicks START: start countdown timer overlay on page
14. Display focus score (starts at 100) next to timer
15. Confirm timer counts down correctly and score displays as 100

Phase 4 — AI Interventions (Day 5)

Goal: When DISTRACTED fires, AI message appears as overlay within 2 seconds.

16. On DISTRACTED event: send intervention prompt to Cerebras with full session context
17. Display AI response as small overlay div (bottom-right corner, non-blocking)
18. Start 30-second timer after AI message is shown

19. If user is still DISTRACTED after 30 seconds: deduct 1 point, increment distraction_count
20. Test: look away → 3 seconds → overlay appears → check score decreases

Phase 5 — Full Escalation Ladder (Days 6–7)

Goal: All 5 escalation levels trigger correctly with AI message always included.

21. Level 2: if distraction_count >= 3, add video skip (video.currentTime -= 30) or window.scrollBy(0, -300)
22. Level 3: if count >= 4, add Web Audio API beep alongside AI message
23. Level 4: if count >= 5, inject CSS border flash (2 seconds) alongside AI message
24. Level 5: if count >= 6, replace overlay with full chatbot conversation panel
25. Test each level in sequence. Confirm AI message appears at every level.

Phase 6 — Post-Session Debrief + Daily Report (Day 8)

Goal: Complete post-session flow works and daily report is generated correctly.

26. When timer reaches 0 or user clicks END: stop detection, show post-session panel
27. Start post-session AI debrief conversation using debrief prompt
28. After debrief: show two rating sliders (learning quality 1–10, focus quality 1–10)
29. Compile daily report JSON from all session data
30. Display report summary to user + provide download link or auto-submit to your data endpoint
31. Test full session start to finish: pre-chat → study → distraction → debrief → report

6. Experiment Design

6.1 Participants

Criterion	Requirement	Reason
Minimum age	18 years old	Adult ADHD, capacity for informed consent
ADHD screening	Score >= 14 on ASRS-v1.1 (Part A)	WHO-validated tool, accepted in academic research
Device	Laptop or desktop with webcam + Chrome browser	Extension compatibility
Language	English (interface and AI responses)	Current system language
Sample size	Minimum 15, target 20 participants	Adequate for a pilot study
Recruitment	Reddit (r/ADHD, r/adhdwomen, r/ADHD_Programmers), Facebook ADHD groups, Discord	Global reach

6.2 ASRS-v1.1 Screening

Every participant completes the first 6 questions of the Adult ADHD Self-Report Scale before enrolling. Participants scoring 14 or above on Part A are included. This is a free, WHO-validated instrument widely used in academic research as a proxy for ADHD diagnosis.

6.3 Experiment Timeline (14 Days)

Days	Phase	System State	Daily Requirement
Day 0	Onboarding	Off	ASRS screening + baseline questionnaire + extension install
Days 1–3	Calibration	Monitoring only — no interventions fired	1 study session minimum
Days 4–13	Main Trial	Full system — randomized conditions	1 study session + daily questionnaire
Day 14	Exit	Off	Final questionnaire + optional exit interview

6.4 Randomized Control Conditions

During the 10 main trial days (Days 4–13), each day is randomly assigned to one of three conditions. The random assignment is generated by the extension and hidden from the user until after the experiment ends.

Condition	Days Assigned	System Behavior
A — Full System	6 out of 10 days	All 5 intervention levels active as designed. Score and timer fully functional.
B — Sham System	2 out of 10 days	System appears fully active (timer, score visible) but interventions are randomly skipped 50% of the time. User cannot tell.
C — No System	2 out of 10 days	Extension silently disabled. Timer and score hidden. User is NOT informed until after the 14 days.

7. Questionnaires

7.1 Pre-Experiment (Day 0 only)

#	Question	Response Type
1	What is your age?	Number
2	What is your gender?	Multiple choice
3	What country are you in?	Text
4	Do you have a formal ADHD diagnosis from a doctor?	Yes / No / Suspected only
5	What ADHD medication do you take, if any?	Text / None
6	What type of content do you most often study? (videos, articles, PDFs, etc.)	Multiple choice
7	On a scale of 1–10, how would you rate your typical ability to focus while studying?	1–10
8	How many hours per day do you typically study?	Number
9	What browser and operating system are you using?	Text
10	Do you wear glasses or contacts?	Yes / No

7.2 Daily Questionnaire (after each session, Days 4–13)

#	Question	Response Type
1	How long was your study session today? (minutes)	Number
2	What was your final focus score today?	Number (auto-filled by extension)
3	On a scale of 1–10, how focused did you feel during this session?	1–10
4	On a scale of 1–10, how much do you feel you actually learned today?	1–10
5	Did you notice the tool intervening today?	Yes / No / Not sure
6	When the tool intervened, did it help you refocus?	Always / Sometimes / Rarely / Never / N/A
7	Which type of intervention felt most helpful today?	AI message / Video skip / Scroll / Sound / Border flash / Full chatbot / None
8	Which intervention felt most annoying or disruptive?	Same options / None

#	Question	Response Type
9	Describe how your focus felt today in 1–2 sentences.	Free text
10	Did you experience any of the following today? (select all that apply)	Restlessness / Zoning out / Racing thoughts / Overwhelm / Hyperfocus / None

7.3 Post-Experiment (Day 14 only)

#	Question	Response Type
1	Compared to before the experiment, has your ability to focus while studying changed?	Much worse / Worse / Same / Better / Much better
2	On a scale of 1–10, how satisfied are you with the tool overall?	1–10
3	Would you continue using this tool after the experiment?	Definitely yes / Probably / Not sure / Probably no / Definitely no
4	What was the single most helpful feature of the tool?	Free text
5	What was the single biggest problem with the tool?	Free text
6	Did the AI messages feel relevant to what you were actually studying?	Always / Sometimes / Rarely / Never
7	Did talking to the AI before your session help you get started?	Yes / Somewhat / No
8	Did the post-session AI debrief feel useful?	Yes / Somewhat / No
9	Did the focus score (100 points) motivate you to stay focused?	Yes / Somewhat / No
10	Any other comments, suggestions, or observations?	Free text

8. Metrics & Analysis Plan

8.1 Quantitative Metrics (from extension logs)

Metric	Definition	Expected Direction
Distraction frequency	DISTRACTED events per hour of study time	Decrease over 14 days
Final focus score	Score out of 100 at end of each session	Increase over 14 days
AI refocus success rate	% of Level 1 interventions with no distraction in next 5 min	Increase over 14 days
Escalation level reached	Average highest level triggered per session	Decrease over 14 days
Session duration	Total minutes per study session	Increase or stabilize
AI vs. Mechanical effectiveness	Refocus rate after Level 1 only vs. Levels 2–5	Direct comparison

8.2 Qualitative Metrics (from questionnaires + AI debrief)

Metric	Source	Analysis Method
Self-rated focus (1–10)	Daily questionnaire Q3	Linear trend over 14 days per participant
Self-rated learning quality (1–10)	Daily questionnaire Q4	Linear trend over 14 days
Intervention preference	Daily questionnaire Q7–8	Frequency distribution per intervention type
ADHD symptom frequency	Daily questionnaire Q10	Count per symptom type over time
Tool satisfaction	Post questionnaire Q2	Descriptive statistics
Continued use intention	Post questionnaire Q3	Percentage breakdown
Open-ended themes	Daily Q9 + Post Q4/5/10	Thematic coding (2 independent coders)

8.3 Condition Comparison (A vs B vs C)

- Primary: Mean distraction frequency on A-days vs B-days vs C-days
- Secondary: Mean self-rated focus score on A-days vs B-days vs C-days
- Statistical test: Repeated measures ANOVA or Wilcoxon signed-rank test (appropriate for n=15-20, non-parametric)

- Effect size: Cohen's d for pairwise comparisons

9. Participant Recruitment

9.1 Target Communities

Platform	Community	Approx. Members	Strategy
Reddit	r/ADHD	1.2 million	Post in weekly thread or create new research post
Reddit	r/adhdwomen	400k	Post research recruitment with empathetic framing
Reddit	r/ADHD_Programmers	80k	Technically inclined users, likely to install extension
Facebook	ADHD Adults Support Group	Variable	Message admin for permission to post research
Discord	ADHD-focused servers	Variable	Search for active servers, ask moderators

9.2 Recruitment Message

```
Title: [Research] AI Chrome Extension to Help ADHD Focus While Studying —
Looking for Testers

Hi everyone,

I'm an undergraduate researcher building a Chrome extension that uses
eye tracking + AI to detect when you lose focus while studying and help
you refocus — without being annoying about it.

The tool:
- Watches your gaze/blink patterns (no video recorded, no data sent to servers)
- Chats with you before/after each session to understand your goals
- Sends personalized AI messages when you get distracted
- Only escalates to louder interventions if the AI messages don't work
- Tracks your focus score (100 points) and sends you a daily report

I need 15-20 volunteers to test it for 14 days.

What's involved:
- Install the Chrome extension (free, no account needed)
- Use it during your normal study sessions for 14 days
- Fill in a short daily form (5 min max)

Requirements:
- 18+ years old
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- ADHD (self-diagnosis OK — you'll take a short 6-question screener)
- Laptop or desktop with webcam + Chrome browser

Your data is fully anonymous.

Interested? DM me or comment below. Happy to answer questions.

— Omar
```

10. Full Project Timeline

Week	Dates	Goal	Deliverable
Week 1	Mar 10–16	Phase 1 & 2: Extension + distraction detection	Eye tracking working in browser
Week 2	Mar 17–23	Phase 3 & 4: Pre-session chat + AI interventions	Full pre-session flow + AI overlay
Week 3	Mar 24–30	Phase 5 & 6: Escalation + post-session debrief + report	Complete end-to-end system
Week 4	Mar 31–Apr 6	Testing + bug fixing on your own machine	Stable v1.0 extension
Week 5	Apr 7–13	Recruit participants, post in communities	15+ confirmed participants
Weeks 6–7	Apr 14–27	Run 14-day experiment	Raw data collected from all participants
Week 8	Apr 28–May 4	Analyze data, generate statistics + charts	Results section complete
Weeks 9–10	May 5–18	Write full paper draft	Complete paper draft
Week 11	May 19–25	Revise, proofread, format for ASSETS 2026	Submission-ready paper
Deadline	June 24	Submit to ASSETS 2026	Submitted

11. Research Paper Outline

Target: 10–12 pages in ACM double-column format (ASSETS 2026).

Section	Content	Approx. Length
Abstract	Problem, system design, experiment method, key finding, implication	150 words
1. Introduction	ADHD impact on learning, gap in existing tools, our contribution, research question	1 page
2. Related Work	Eye tracking in education, ADHD + digital tools, AI behavioral intervention, positioning	1.5 pages
3. System Design	Architecture (3 layers), session lifecycle, AI prompt design, escalation ladder, score system	2 pages
4. Study Design	Participants, ASRS screening, 14-day protocol, 3 conditions, measures, analysis plan	1.5 pages
5. Results	Distraction trends, score trends, condition comparison, AI vs. mechanical effectiveness, qualitative themes	2 pages
6. Discussion	Key findings, did AI outperform mechanical?, limitations, implications for ADHD tech design	1.5 pages
7. Conclusion	Summary, future work (clinical study, mobile version)	0.5 pages
References	20–30 citations in ACM format	1 page

12. Ethical Considerations

Concern	How It Is Addressed
Informed consent	Plain-language description sent before enrollment. Participants can withdraw at any time without penalty.
Data privacy	No video or audio recorded. Eye tracking processed locally. Only aggregated statistics stored remotely.
ADHD screening	ASRS-v1.1 used as inclusion criterion. Self-diagnosis accepted for pilot study — clearly stated in paper's limitations section.
Placebo effect	Randomized sham condition (B) and silent-off condition (C) control for placebo and expectation effects.
Participant burden	Daily questionnaire limited to 5 minutes. Participants may skip days without losing their place in the study.
Data anonymity	No names or personal identifiers collected. Each participant assigned a random ID code at enrollment.
Deception (Condition C)	Participants are informed before enrollment that some days the system may behave differently. Full debrief provided on Day 14.

Start with Phase 1 tomorrow. The paper writes itself once you have real data. Every day you delay the build is a day lost from your experiment window.

Good luck, Omar.